ELSEVIER

Ecological Modelling 108 (1998) 265–276

ECOLOGICAL MODELLING

Model of the total exchange carbon flux for terrestrial ecosystems

Leonid L. Golubyatnikov [a,*], Elizaveta A. Denisenko [b], Yuri M. Svirezhev [c]

[a] *Laboratory of Mathematical Ecology, Institute of Atmospheric Physics, Russian Academy of Sciences, Pyzhevsky 3, Moscow, 109017, Russia*

[b] *Institute of Geography, Russian Academy of Sciences, Staromonetny 29, Moscow, 109017, Russia*

[c] *Potsdam Institute for Climate Impact Research, P.O.Box 601203, 14412 Potsdam, Germany*

Accepted 21 May 1997

Abstract

We present here an approach to evaluation of the total exchange carbon flux by means of mathematical modelling. This flux is defined as a difference between carbon income into the net primary production and its emission from dead organic matter due to both biological and non-biological processes. We suggest the model which enables us to determine functioning of forest and grass ecosystems as carbon sources or carbon sinks and to evaluate intensity of these sources and sinks. The model is applied for the European territory of Russia as a case study. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Mathematical model; Carbon flux; Terrestrial ecosystem

1. Introduction

Carbon plays an important role in many environmental processes. It is the great factor of coevolution of the biosphere and anthroposphere. The impact of increasing atmospheric carbon dioxide concentration on global climate is an environmental issue of human activity. Today, when mankind has an increasing impact on the biosphere, the balance evaluation for natural carbon fluxes is one of the main problems.

Obviously, at each point of the land surface there is a local carbon cycle of the 'atmosphere–vegetation–soil' type. All the local cycles are combined into the global cycle by the atmosphere, in which the time of carbon intermixing totals to 3–4 months (Schimel, 1995). Inhomogeneities in the global vegetation pattern and climatic conditions generate those among local carbon cycles. The atmosphere is a natural integrator for the work done by local cycles and its dynamic func-

* Corresponding author. Tel.: +7 95 2315565; fax: +7 95 2331652/9531652; e-mail: bbm@dionis.iasnet.ru

0304-3800/98/$19.00
PII S0304-3800(98)00033-7

tion is, in a sense, to average the local inhomogeneities of the global carbon cycle.

At a particular time moment each point of the earth's surface can be:

- a carbon source (carbon release into atmosphere is higher than absorption);
- a carbon sink (carbon release into atmosphere is smaller than absorption);
- an equilibrium (the local carbon fluxes are balanced).

The aim of our research is to develop a method which would permit to determine functioning of forest and grass ecosystems as sources or sinks for carbon and to evaluate intensity of these sources and sinks for a given territory. Unfortunately, to solve this problem we can not use results of direct measurements for the necessary carbon fluxes, because those results have huge variability in space and time and they are not sufficient for making extrapolation at a regional level.

We try to solve the problem by means of modelling the total exchange carbon flux for the terrestrial territory under study.

2. Methods and discussion

A kind of indirect methodology is suggested to solve the problem. It uses:

- a spatially distributed dynamic model of the global carbon cycle;
- estimations for the annual net primary production (NPP) and phytomass data for various vegetation patterns;
- the information about natural and agricultural lands on a given territory;
- the map of soil carbon storage and the renewal rates data for soil carbon;
- data on decomposition rates for slowly decaying organic substances.

2.1. Problem formalization

Let us consider terrestrial part of the global carbon cycle as a partially isolated dynamic system. We consider the following model variables: $C(t)$ is the amount of atmospheric carbon at time moment t, $B(x, y, t)$ is the density of living phytomass at a point (x, y) in carbon units, $D(x, y, t)$ is the density of dead organic matter in carbon units. Both dead phytomass (mortmass) and organic matter in soil (humus) are included into the dead organic matter. Note that our knowledge about real nonlinear feed-back mechanisms for biosphere machine are uncertain. We assume a linear description both for living phytomass and dead organic matter as the simplest mathematical hypothesis.

The local equation for the density of living phytomass $B(x, y, t)$ takes on the following form:

$$\partial B/\partial t = P(x, y, t) - mB \quad (1)$$

where $P(x, y, t)$ is the annual NPP at a point (x, y), $m(x, y, t) = 1/\tau$, where τ is the residence time for carbon in the living phytomass. Due to use of the annual NPP, the characteristic time in the model is chosen one year.

The local equation for the density of dead organic matter $D(x, y, t)$ takes on the following form:

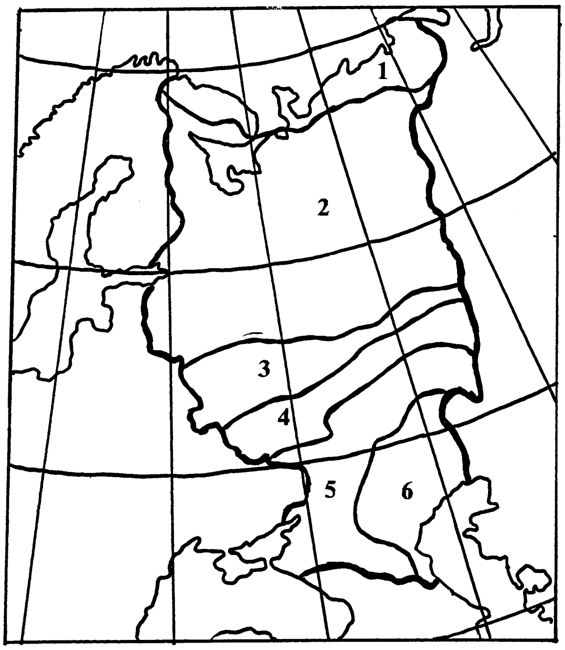


Fig. 1. Ecosystems of the European territory of Russia: 1, Tundra; 2, Taiga forest; 3, Broadleaved forest; 4, Meadow steppe; 5, True steppe; 6, Semi-desert.

Table 1
Estimations of net primary production (NPP), nitrogen concentrations within living phytomass (N) and specific carbon concentration (SCC) for different ecosystem types

Ecosystem type	Area (10^6 km^2)	NPP (kg/m^2 per year)	N (g/m^2)	SCC
Tundra	0.11	0.2	5.3	0.56
Taiga forest	2.00	0.8	6.7	0.40
Broadleaved forest	0.31	1.1	11.4	0.53
Meadow steppe	0.47	1.8	12.7	0.37
True steppe	0.43	1.3	10.9	0.38
Semi-desert	0.30	0.12	1.8	0.45
Weighted average	—	0.94	8.0	—

$$\partial B/\partial t = mB - \delta D - qD \quad (2)$$

where $\delta(x, y, t)$ is a part of the dead organic matter which is decayed annually and is emitted into the atmosphere as CO_2, $q(x, y, t)$ is a part of the dead organic matter which is lost annually for ecosystems (for example, with runoff, dusty storms, soil erosion, etc.). The functions $\delta(x, y, t)$ and $q(x, y, t)$ can be represented by the following sums:

$$\delta(x, y, t) = \delta_m(x, y, t) + \delta_s(x, y, t)$$

$$q(x, y, t) = q_m(x, y, t) + q_s(x, y, t)$$

where δ_m, δ_s are parts of dead phytomass (mortmass) and organic matter in soil (humus), respectively, which is decayed annually and is emitted into the atmosphere as CO_2, q_m, q_s are parts of mortmass and humus respectively which annual is lost for ecosystems due to non biological processes.

The equation for the atmospheric carbon $C(t)$:

$$\mathrm{d}C/\mathrm{d}t = -\int_\Omega P(x, y, t)\,\mathrm{d}x\,\mathrm{d}y + Q_{CO} + e(t) + \int_\Omega \delta D\,\mathrm{d}x\,\mathrm{d}y \quad (3)$$

where Q_{CO} is the exchange flux between the atmosphere, terrestrial biota and the ocean, $e(t)$ is the anthropogenic emission of carbon, Ω is the terrestrial area.

According to the spatially distributed model of the global carbon cycle (1)–(3) there are three carbon fluxes at each point (x, y) of the land surface: $P(x, y, z)$ is the flux from the atmosphere into phytomass, $\{\delta_m + \delta_s\}D(x, y, t)$ is the flux from the dead organic matter into the atmosphere, $\{q_m + q_s\}D(x, y, t)$ is the abiotic loss from ecosystem. Then the total exchange carbon flux we can represented by:

$$V(x, y, t) = P(x, y, t) - \{\delta_m + \delta_s + q_m + q_s\}D(x, y, t),$$

that is as the difference between carbon income into the annual NPP and its emission from dead organic matter due to both biological and non-biological processes per year.

Therefore if $V(x, y, t) > 0$, then a given point (x, y) at a particular time moment t is a carbon sink, if $V(x, y, t) < 0$, then a given point (x, y) at a particular time t is a carbon source.

Thus the aim of our research is to calculate the value of $V(x, y, t)$ for current time moment t and the forest–grass ecosystems of a given territory.

2.2. Calibration of model

The strong assumption that any local carbon cycle is in steady state—the so-called steady-state hypothesis—was used in calibrating many existing dynamic models of the global carbon cycle with spatial distributions. In other words, it was assumed that at some moment, which is referred to as 'preindustrial world', all the local carbon cycles were balanced. For example, this hypothesis was used in calibrating spatial patterns of the global carbon cycle created within the framework of such well-known models as the Moscow Global Biosphere Model (Krapivin et al., 1982; Svirezhev

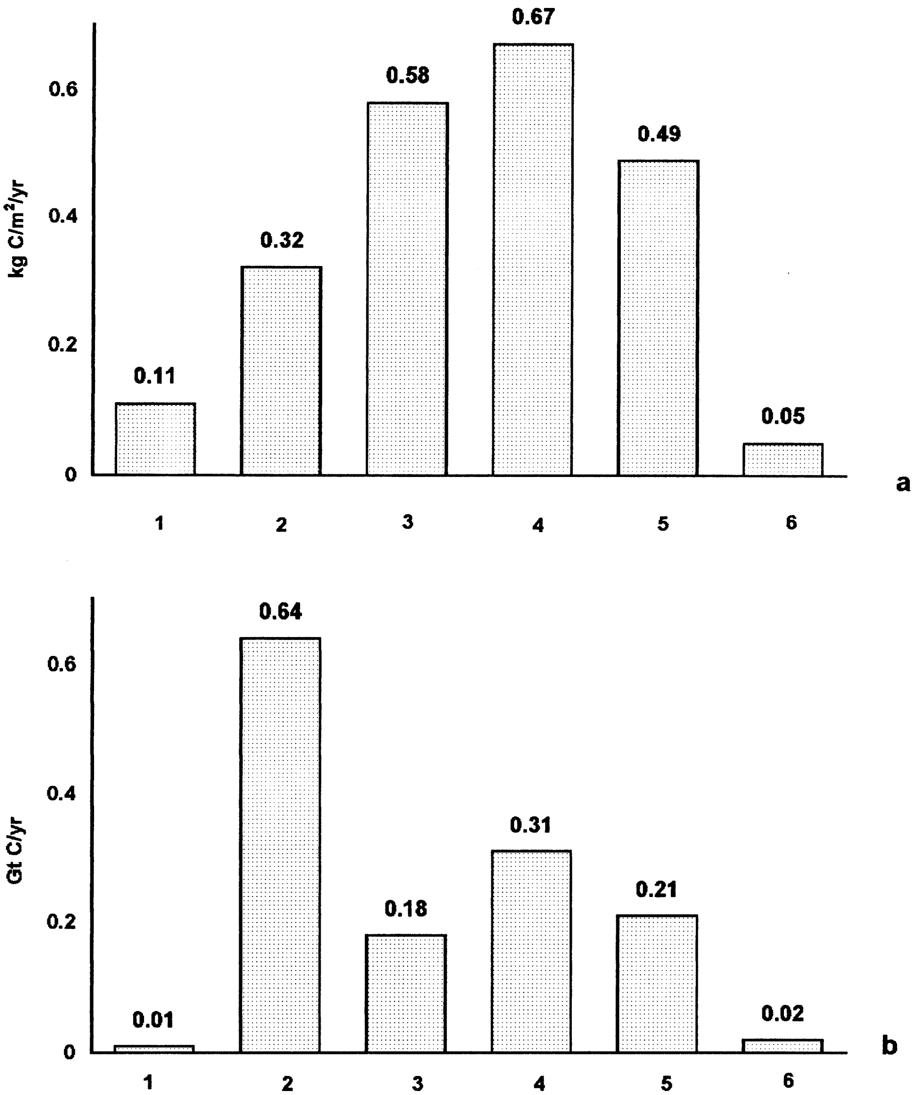


Fig. 2. Estimations of annual carbon fluxes from the atmosphere into phytomass for ecosystems of ETR ((a), kg C/m^2 per year; (b), Gt C/year). Notations are the same as for Fig. 1.

et al., 1985; Krapivin, 1993), the Frankfurt Biosphere Model (Ludeke et al., 1994).

In our method, on the contrary, the steady-state hypothesis is rejected. To calibrate our model we have used extensive experimental information. The model is applied for the European territory of Russia (ETR) as a case study. The European part of Russia is a vast territory with area ca. 3.6 million km^2 that is located between 30° and 60° eastern longitude and 45° and 70° northern latitude. We considered six ecosystems on the given territory: tundra, taiga forest, broadleaved forest, meadow steppe, true steppe and semi-desert (Fig. 1). This subdivision is based on using a land formation (biome) as a unit for territory classification. Note that taiga forest ecosystem occupies more than one-half of the territory under concern.

To calculate the carbon flux into the annual NPP we have used empirical data about NPP

Table 2
Characteristics of the upper meter layer of soil for the ETR

Soil type	Area (10^3 km^2)	Carbon store		Renewal of the carbon per year	
		kg/m^2	Gt	g/m^2	Mt
Podbur	57	20.7	1.2	59.5	3.4
Gley tundra soil	107	17.4	1.9	32.4	3.5
Peatbog soils of northern taiga	85	37.0	3.1	39.1	3.3
Gley and gley-podzolic soil	204	10.7	2.2	14.8	3.0
Podzols of northern taiga	182	10.0	1.8	22.4	4.1
Hydromorphic podzolic soils	208	37.0	7.7	120.9	25.1
Podzolic	319	9.1	2.9	12.9	4.1
Podzols of the middle taiga	139	10.5	1.4	15.6	2.2
Peatbog soils of southern taiga	34	63.8	2.2	147.3	5.0
Sodpodzolic loamy	636	7.4	4.7	18.5	11.8
Sodpodzolic sandy	233	5.9	1.4	29.9	7.0
Gray forest soils	240	15.8	3.8	11.8	2.8
Leached chernozems	233	30.3	7.1	10.0	2.3
Typic chernozems	91	31.9	2.9	13.0	1.2
Ordinary chernozems	117	26.0	3.0	13.6	1.6
Meadow chernozem-like	30	33.4	1.0	7.5	0.2
Southern chernozems	125	19.3	2.4	26.0	3.3
Chestnut soils	208	10.3	2.1	5.3	1.1
Chernozemic calcareous soils	108	23.9	2.6	8.8	1.0
Semidesert soils	258	2.9	0.7	0.6	0.2
Weighted average	—	15.6	—	23.8	—
Total	3613	—	56.2	—	86.1

(Bazilevich et al., 1986; Bazilevich, 1993) (Table 1). Meadow steppe ecosystem has the maximal NPP, which smoothly decrease to the south and north of it. Note that tundra, taiga forest and broadleaved forest ecosystems, the overground NPP is higher than the underground NPP. Steppe and semi-desert ecosystems, the underground NPP is higher than the overground NPP.

Using chemical analysis data about nitrogen concentrations collected by Rodin and Bazilevich (1965) and C/N ratios within phytomass for given ecosystems (Bazilevich and Titlyanova, 1978), we have calculated specific carbon concentrations (Table 1). For tundra and broadleaved forest ecosystems, it is higher than 50%, for taiga and steppe ecosystems it is about 40%. Then we have expressed NPP experimental data in the carbon units.

Thus we can estimate annual carbon fluxes from the atmosphere into phytomass for ecosystems of ETR (Fig. 2a,b). Semi-desert ecosystem has the minimal flux on a unit of area. The maximal magnitude for this flux on a unit of area is observed at meadow steppe ecosystem. The mean value for ETR equals ca. 290 g C/m^2. For ecosystems of ETR we have obtained the following estimation for this flux: every year taiga absorbs about 0.6; meadow steppe, about 0.3; true steppe and broadleaved forest ecosystems, about 0.2 and tundra and semi-desert ecosystems absorb smaller than 0.02 Gt of carbon. Total annual carbon flux from the atmosphere into phytomass for ETR is 1.4 Gt.

Each ecosystem loses a part of the dead phytomass carbon due to non biological processes, for example, with runoff, dusty storms, soil erosion, etc. Note that only average values of these losses are available. According to Kovda (1975) tundra ecosystem has the maximal value of the abiotic losses of carbon among the ecosystem types. Every year it is about 6% NPP. For taiga ecosystem this magnitude is about 5%. For an-

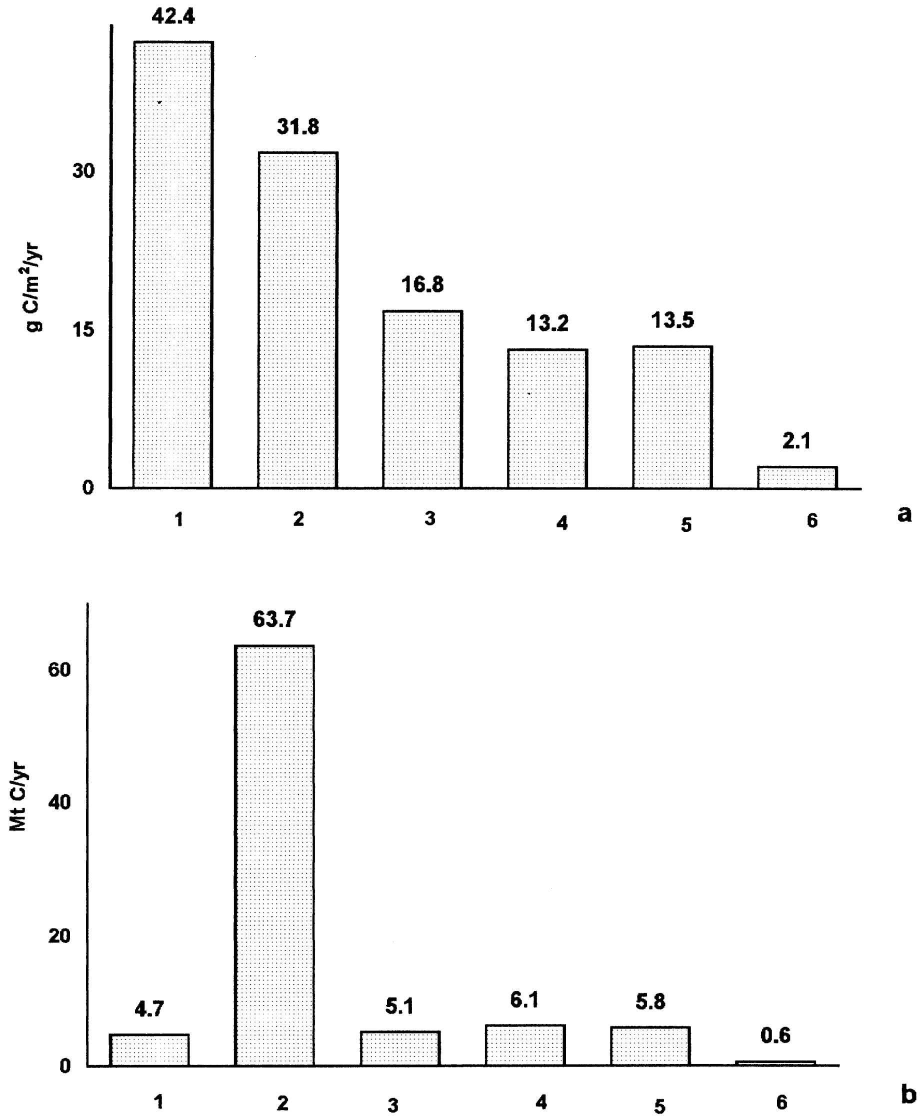


Fig. 3. Estimations of annual carbon fluxes from organic matter in soil for ecosystems of ETR ((a), g C/m^2 per year; (b), Mt C/year). Notations are the same as for Fig. 1.

other ecosystems of ETR this magnitude does not exceed 1%. These values are the estimates for annual abiotic loss carbon fluxes from the dead phytomass.

In order to estimate the annual carbon flux from organic matter in soil we have used the data about renewal of the soil carbon for ETR obtained by Brovkin et al. (1995). They accounted the spatial distribution of soil carbon storage and, using the radiocarbon method, designed the model describing carbon isotopes dynamics in soil (Cherkinsky and Brovkin, 1993) and estimated the renewal rates of soil carbon for different soil types (Table 2).

According to the results by these authors the carbon storage is about 20 kg/m^2 at the areas of podburs and gley tundra soils. It decreases to 10 kg/m^2 for podzolic and sodpodzolic soils. Cher-

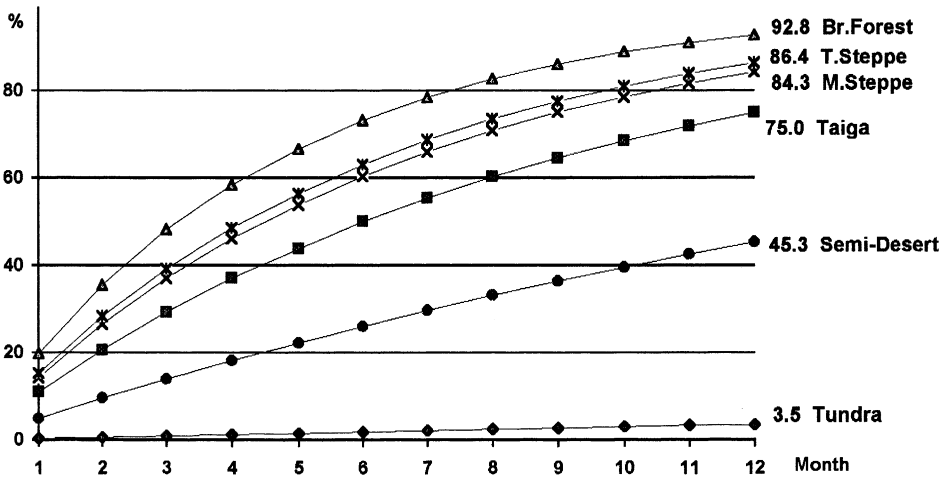


Fig. 4. Estimation of annual decomposition rate (%) for slowly decaying organic substances.

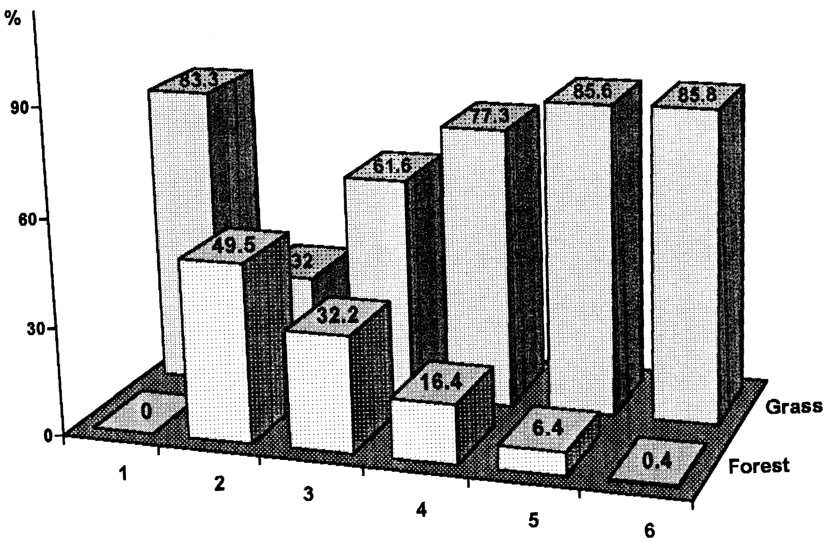


Fig. 5. Area parts for forest and grass ecosystems of ETR (%).Notations are the same as for Fig. 1.

nozem fertile soils are characterized by the higher values of carbon content in humus (20–35 kg/m^2). Chestnut and semidesert soils in the south-eastern region of ETR with dry and hot climate condition have a low level of carbon content (3–10 kg/m^2).

Northern soils are characterized by significant renewal rates of soil carbon (33–60 g/m^2 per year). Carbon in southern peat bogs (middle and southern taiga) have about two times more renewal rates than in northern region (120–150 g/m^2 per year). Carbon in chernozem soils renews at a low level (9–13 g/m^2 per year). It is interesting that chernozem fertile soils contain much carbon in humus and have the low renewal level for soil carbon.

The magnitude of renewal for the soil carbon characterizes the exchange carbon flux in the soil organic matter. Hence, we can estimate annual carbon fluxes from organic matter in soil (humus) for ecosystems of ETR (Fig. 3a,b). Tundra

Table 3
Chemical composition of mortmass

	% Dry matter				
Organisms	Cinder	Albuminous substances	Carbohydrates	Lignin	Lipids, tannins
Lichens	2–6	3–5	65–90	8–10	1–3
Mosses	3–10	5–10	45–85	–	5–10
Coniferous					
Wood	0.1–1	0.5–1	60–75	25–30	2–12
Needle	2–5	3–8	30–40	20–30	15–20
Deciduous					
Wood	0.1–1	0.5–1	60–80	20–25	5–15
Leaves	3–8	4–10	25–45	20–30	5–15
Perennial herbage					
Grass	5–10	5–15	50–75	15–20	2–10
Leguminous plants	5–10	10–20	40–55	15–20	2–10

ecosystem has the maximal flux from a unit of area. The minimal magnitude for this flux from a unit of area is observed at semi-desert ecosystem. The mean value for ETR equals ca. 24 g C/m^2. For ecosystems of ETR we have obtained the following estimation for this flux: every year taiga ecosystem soils lose about 64 Mt of carbon, tundra soils and broadleaved forest soils lose about 5 Mt of carbon, steppe ecosystems soils lose about 6 Mt of carbon, semi-desert ecosystem soil loses smaller than 1 Mt of carbon. Total annual carbon flux from organic matter in soil for ETR is 86 Mt.

It is known that NPP is transformed through trophic chains: about 50% for meadow and true steppe, 30–37% for tundra, taiga forest, broadleaved forest, semi-desert ecosystems (Gilmanov, 1978).

The part of the annual increment that is not fixed in perennial plant organs and is not consumed has many slowly decaying substances, such as lignin, cellulose. In order to calculate the annual decomposition rate for slowly decaying organic substances, we have used cellulose decomposition experimental data for the different ecosystems and have assumed the exponential hypothesis for the decomposition rate dynamics. According to experimental data (Zlotin and Chukanova, 1973; Bazilevich et al., 1993) in the warm season the monthly cellulose decomposition rate is about 20% for the broadleaved forest ecosystem, 11–15% for taiga and steppe ecosystems, about 5% for the semi-desert ecosystem and 0.3% for the tundra ecosystem. Having assumed the exponential hypothesis, we have obtained the following estimates of annual decomposition rate for slowly decaying organic substances (Fig. 4): the maximal magnitude for the rate is observed at broadleaved forest ecosystem (ca. 93%), where climatic conditions are optimal for destructive processes, the minimal magnitude for the rate is observed at tundra ecosystem (ca. 3.5%), where low temperature does not permit micro-organisms to decay organic matter. Note that the annual destruction values for cellulose at taiga and broadleaved forest ecosystems are known from the field experiments of other authors (Grishina et al., 1990; Bazilevich et al., 1993). According to those experimental data, about 77 and 93% of cellulose are decayed at taiga and broadleaved forest ecosystems per year, respectively. Thus our estimates of the annual decomposition rate for slowly decaying organic substances coincide very much with the field experiments results.

In our model we considered forest and grass ecosystems. Urbanized and technogenic territories were excluded from our analysis. Agroeco-systems were included into grass ecosystems. Using the afforestation map and the information about natural and agricultural lands for the given territory (Egorov, 1975) we have calculated area parts for forest and grass ecosystems of ETR (Fig. 5). These diagrams permit to receive general repre-

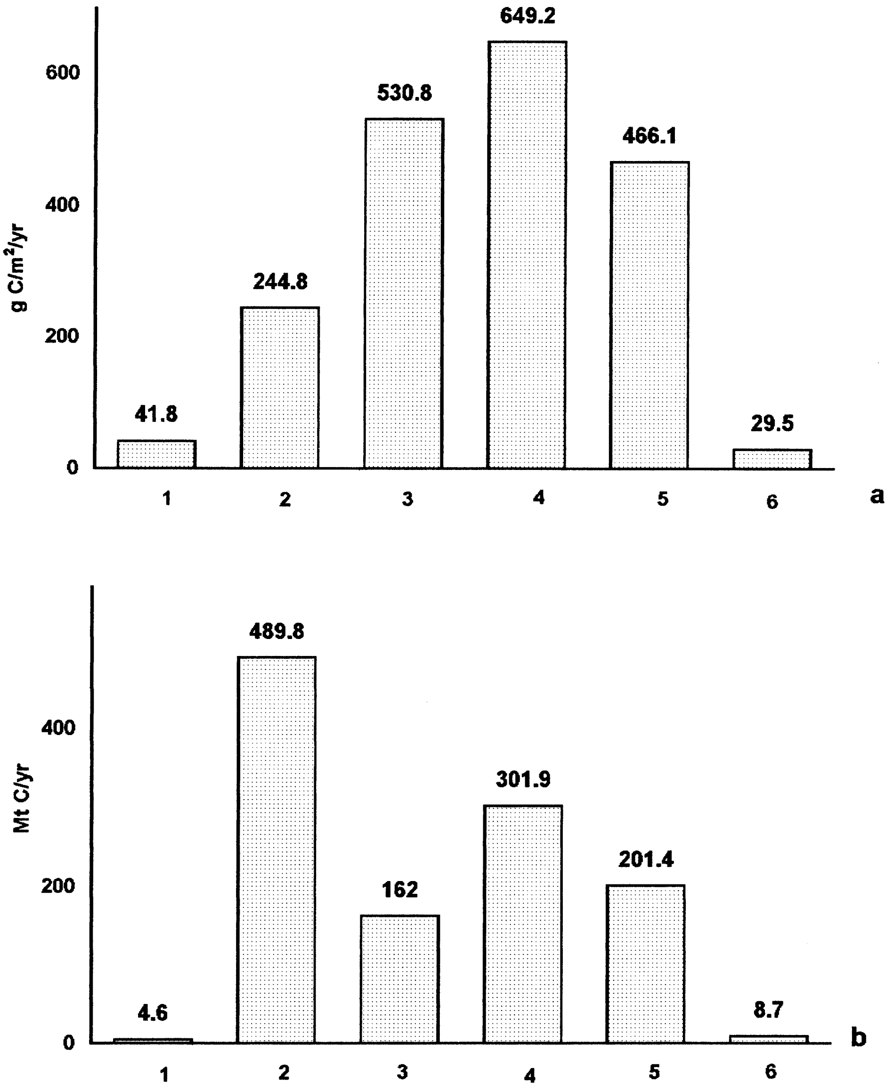


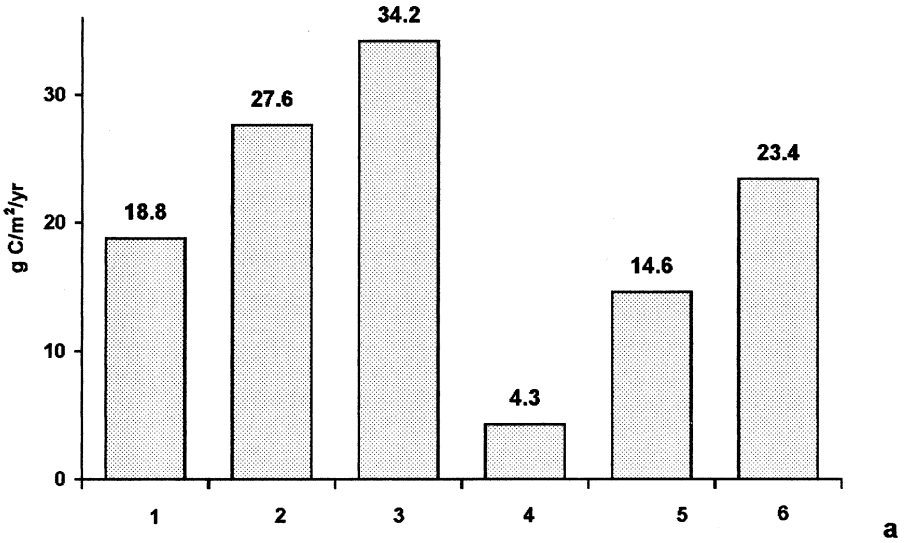
Fig. 6. Estimations of annual carbon fluxes from dead phytomass for ecosystems of ETR ((a), g C/m^2 per year; (b), Mt C/year). Notations are the same as for Fig. 1.

sentation about spatial distribution for forest and grass ecosystems on ETR.

Let us denote by a_f and a_g the carbon flux from dead phytomass ($\delta_m D$) for forest and grass ecosystems, respectively, and denote by p_f and p_g the parts of forest and grass ecosystems, respectively, at each biome on the given territory. Then for each point (x, y) the mean value for the carbon flux from dead phytomass to the atmosphere can be calculated as:

$$\delta_m D = a_f p_f + a_g p_g$$

Having used data about the annual increment transformation through trophic chains, chemical composition data for mortmass (Alexandrova, 1980) (see Table 3), and the annual decomposition rate for slowly decaying organic substances we can estimate the annual carbon flux from dead phytomass (Fig. 6a,b). Meadow steppe ecosystem has the maximal flux from a unit of area. Note



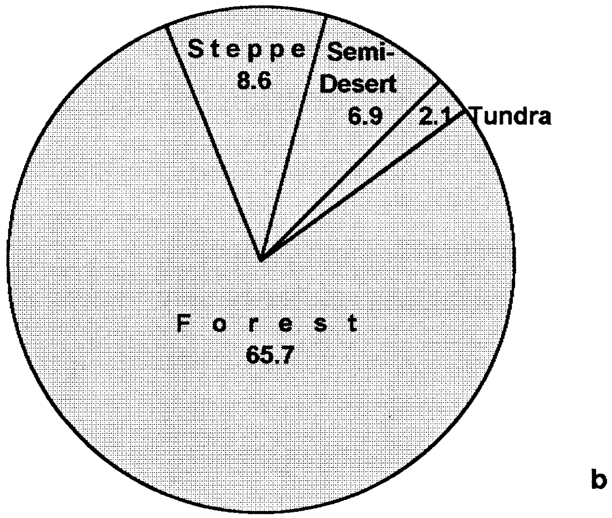


Fig. 7. Estimations of annual total exchange carbon flux for ETR ecosystems for ecosystems of ETR ((a), g C/m^2 per year; (b), Mt C/year). Notations are the same as for Fig. 1.

that the underground dead phytomass is higher than the overground mortmass for this ecosystem. The minimal magnitude for this flux from a unit of area is observed at semi-desert ecosystem. The mean value of the flux from dead phytomass for ETR equals ca. 323 g C/m^2. For ecosystems of ETR we have obtained the following estimation for this flux: every year taiga ecosystem emits about 490 Mt of carbon, broadleaved forest and steppe emits 160–300 Mt, semi-desert ecosystem about 9 Mt and tundra about 5 Mt of carbon. The total annual carbon flux from dead phytomass for ETR is 1.2 Gt.

Thus we can calculate all the fluxes which are necessary to compute annual total exchange carbon flux according to the dynamic model of spatially distributed global carbon cycle.

3. Results

The simulated significances of the annual total exchange carbon flux for ETR are shown in Fig. 7a,b. Mean value of this flux for ecosystems under consideration equals ca. 22 g C/m^2 per year. The maximal magnitude of the annual total exchange carbon flux is characteristic for broadleaved forest ecosystem (about 34 g C/m^2), the minimal magnitude is characteristic for meadow steppe ecosystem (about 4 g C/m^2).

The magnitude of the total exchange carbon flux can characterize the global role of vegetation communities as carbon accumulators. We have found that a significant role belongs forest ecosystems of ETR, which deposit about 66 Mt carbon every year. This value is in agreement with the estimates obtained by other authors for forest ecosystems of ETR (Kolchugina and Vinson, 1993a,b; Kokorin, 1996). According to these authors, forest ecosystems of ETR absorb 15–90 Mt carbon per year. We have found that steppe ecosystem absorbs up to 9 Mt carbon every year. At semi-desert ecosystem carbon sink magnitude decreases up to 7 Mt. Tundra ecosystem is characterized by the minimal difference between carbon absorption and its emission into the atmosphere, about 2 Mt. The annual total exchange carbon flux for ETR is evaluated as 81 Mt.

Our research has shown that ETR is a carbon sink at present time. The estimates obtained are in agreement with those of other authors (Isaev et al., 1993; Zavarzin, 1994 and others) that ETR is a sink for carbon.

4. Conclusion

In our model for the estimation of the total exchange carbon flux we do not use the hypothesis of steady state for all local carbon cycles. The model is based on the extensive experimental information.

The suggested model permits us to characterize a particular terrestrial territory as a source or a sink for carbon. Certainly, the model cannot take into account the whole set of productive and destructive processes for organic matter. Therefore the results obtained represent some estimates but they nevertheless give an idea of the qualitative pattern of the carbon balance for the territory under study.

The method proposed enables us to determine the state of the total exchange carbon flux at the end of the 20th century. The results can serve as a basis to monitor any further changes in the parameters of this flux.

Acknowledgements

We thank Professor Dmitrii O. Logofet for helpful discussions. We are also deeply appreciated anonymous reviewers for useful comments on the draft version of this manuscript.

References

Alexandrova, L.N., 1980. Soil Organic Matter and Processes of its Transformation. Nauka, Leningrad, p. 227, (in Russian).

Bazilevich, N.I., 1993. Biological Productivity for Ecosystems of Northern Eurasia. Nauka, Moscow, p. 293, (in Russian).

Bazilevich, N.I., Drozdov, A.V., Zlotin, R.I., 1993. Geographical peculiarities of the production–destruction-processes in Northern Eurasia. In: Breymeyer, A. (Ed.), Geography of Organic Matter Production and Decay. PAN IG&PZ, Warsaw, pp. 51–79.

Bazilevich, N.I., Grebenschikov, O.S., Tishkov, A.A., 1986. Geographical Laws of Ecosystem Structure and Ecosystem Functioning. Nauka, Moscow, p. 297, (in Russian).

Bazilevich, N.I., Titlyanova, A.A., 1978. Comparative studies of ecosystem function. In: Breymeyer, A.J., Van Dyne, G.M. (Eds.), International Biological Programme 19. Cambridge University Press, Cambridge, pp. 713–758.

Brovkin, V.A., Cherkinsky, A.E., Goryachkin, S.V., 1995. Estimation and mapping of renovation rate of soil carbon based on the radiocarbon data (European Russia case study). Ecol. Model. (submitted).

Cherkinsky, A.E., Brovkin, V.A., 1993. Dynamics of radiocarbon in soils. Radiocarbon 35, 363–367.

Egorov, V.V. (Ed.), 1975. Natural and Agricultural Lands USSR. Kolos, Moscow, p. 256, (in Russian).

Gilmanov, T.G., 1978. Mathematical Modelling of Biochemical Cycles in Grass Ecosystems. Moscow State University, Moscow, p. 169, (in Russian).

Grishina, L.A., Koptsik, G.N., Makarov, M.I., 1990. Transformation of Soil Organic Substances. Moscow State University, Moscow, p. 88, (in Russian).

Isaev, A.S., Korovin, G.N., Utkin, A.I. et al. 1993. The estimation of the carbon storage and the annual carbon deposition in phytomass for the forest ecosystems in Russia. Lesovedenie, 5, 3–10, (in Russian).

Kokorin, A.O., 1996. The analysis and experimental estimation of boreal forest growth parameters under warming. In: Izrael, Yu.A. (Ed.), Problems of Ecological Monitoring and Ecosystem Modelling, vol. XVI. Gidrometeoizdat, St. Petersburg, pp.72–84, (in Russian).

Kolchugina, T.P., Vinson, T.S., 1993a. Comparison of two methods to assess the carbon budget of forest biomes in the former Soviet Union. Water, Air, Soil Pollut. 70, 207–222.

Kolchugina, T.P., Vinson, T.S., 1993b. Carbon sources and sinks in the of forest biomes in the former Soviet Union. Glob. Biogeochem. Cycl. 7, 291–304.

Kovda V.A., 1975. Biogeochemical Cycles in Nature and their Study. Nauka, Moscow, p. 74, (in Russian).

Krapivin, V.F., 1993. Mathematical model for global ecological investigations. Ecol. Model. 67, 103–127.

Krapivin, V.F., Svirezhev, Yu.M, Tarko, A.M., 1982. Mathematical modelling of the Global Biosphere Processes. Nauka, Moscow, p. 272. (in Russian).

Ludeke, M.K.B., Badeck, F.-W., Otto, R.D., 1994. The Frankfurt Biosphere model: a global process-oriented model for the seasonal and long-term CO_2 exchange between terrestrial ecosystems and the atmosphere. Clim. Res. 4, 143–166.

Rodin, L.E., Bazilevich, N.I., 1965. Organic Matter Dynamics and Biological Circulation of Cinder Elements and Nitrogen in Main Vegetation Types of the Earth. Nauka, Moscow, p. 253, (in Russian).

Svirezhev, Yu.M., Krapivin, V.F., Tarko, A.M., 1985. Modeling of the main biosphere cycles. In: Malone, T.F., Roederer, J.G. (Eds.), Global Change. Cambridge University Press, Cambridge, pp. 298–313.

Schimel, D.S., 1995. Terrestrial ecosystems and the carbon cycle. Glob. Chang. Biol. 1, 77–91.

Zavarzin, G.A., 1994. Carbon cycle for the Natural Ecosystems of Russia. Priroda 7, 15–18, (in Russian).

Zlotin R.I, Chukanova, A.V., 1973. Decomposition of a cellulose as a intensity indicator for microbiological processes into forest–grass ecosystems. In: Snytko, V.A., (Ed), Topological Features of the Matter Behaviour in Ecosystems. Academiya Nauk, Irkutsk, pp. 213–216, (in Russian).